



Jordan University of Science and Technology
Faculty of Science & Arts
Biotechnology & Genetic Engineering Department

IMMUNOLOGY & SEROLOGY LABORATORY (BT336)

First Semester 2023 - 2024

About This Class

The Immunology and Serology Laboratory involves imparting students with hands-on experience in fundamental immunology and serology concepts, emphasizing laboratory techniques such as ELISA, Western blotting, and immunofluorescence. The curriculum covers safety protocols, diagnostic testing for infectious and autoimmune diseases, blood typing, and the use of specialized equipment. Students learn to analyze and interpret results, apply quality control measures, and connect theoretical knowledge to real-life scenarios through case studies. Emphasis is also placed on ethical considerations, and communication skills, preparing students for careers in medical laboratory science and related fields where they play a vital role in clinical diagnostics and research within the realm of immunology and serology.

Course Objectives

The main objectives of the course are:

1. Gain a comprehensive understanding of the fundamental principles of immunology, including the components of the immune system, immune responses, and the role of antibodies and antigens in immune reactions.
2. Develop proficiency in performing essential laboratory techniques used in immunology and serology, such as ELISA, Western blotting, immunofluorescence, precipitation tests, and agglutination tests.
3. Emphasize adherence to laboratory safety protocols, including proper handling of biological samples and the use of personal protective equipment.
4. Learn and apply diagnostic testing skills for the detection of antibodies, antigens, and other immune-related markers, with a focus on diagnosing infectious diseases, autoimmune disorders, and blood typing.

Course Topics

Week	Experiment
(1) 2 - 6/3	Animal Handling and Immunization
(2) 9 - 13/3	Bleeding of Animals and Serum Preparation
(3) 16 - 20/3	White Blood Cells Counting and Differentiation
(4) 23 - 27/3	Isolation of Lymphocytes and Immunoglobulins
(5) 6 - 10/4	Double Immunodiffusion and Single Radial Immunodiffusion Technique (Mancini Test)
(6) 13 - 17/4	Immunoelectrophoresis
(7) 20 - 24/4	Direct and Indirect Immunoagglutination
(8) 27 - 30/4	Enzyme-Linked Immunosorbent Assay (ELISA)
(9) 4 - 8/5	Western Blot
(10) 11 - 15/5	Fluorescent Immunoassay and Radioimmunoassay

Assessment Policy

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Midterm Exam	40 points
Final Exam	40 points
Practical Quiz	10 points
Practical work	10 points
Total	100 points

Course Instructor

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